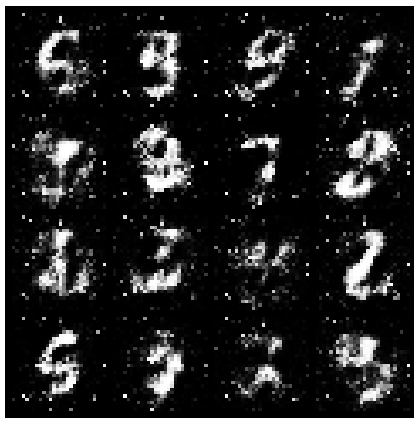




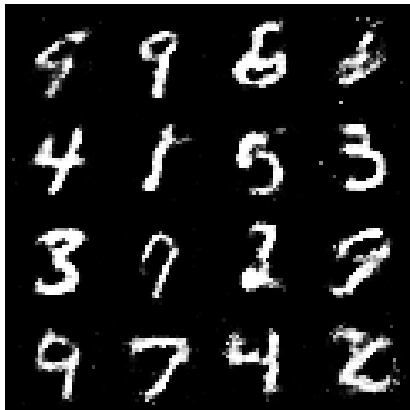
Epoch 5: Early-stage generation with visible noise and incomplete digit structures.

D Loss = 0.9196 | G Loss = 1.1592



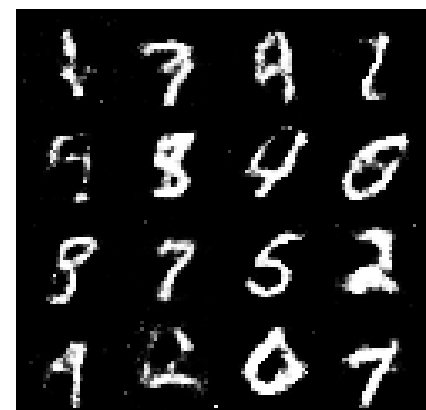
Epoch 15: Digit shapes become more recognizable with improved stroke continuity.

D Loss = 1.0136 | G Loss = 0.7308



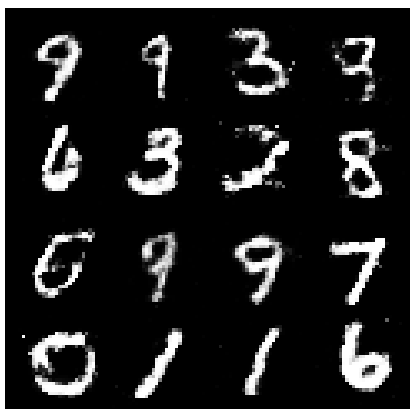
Epoch 30: Generator produces clearer digit diversity with reduced visual distortion.

D Loss = 1.3558 | G Loss = 1.4153



Epoch 45: Stable handwritten digit generation with high readability across samples.

D Loss = 1.1597 | G Loss = 0.7803

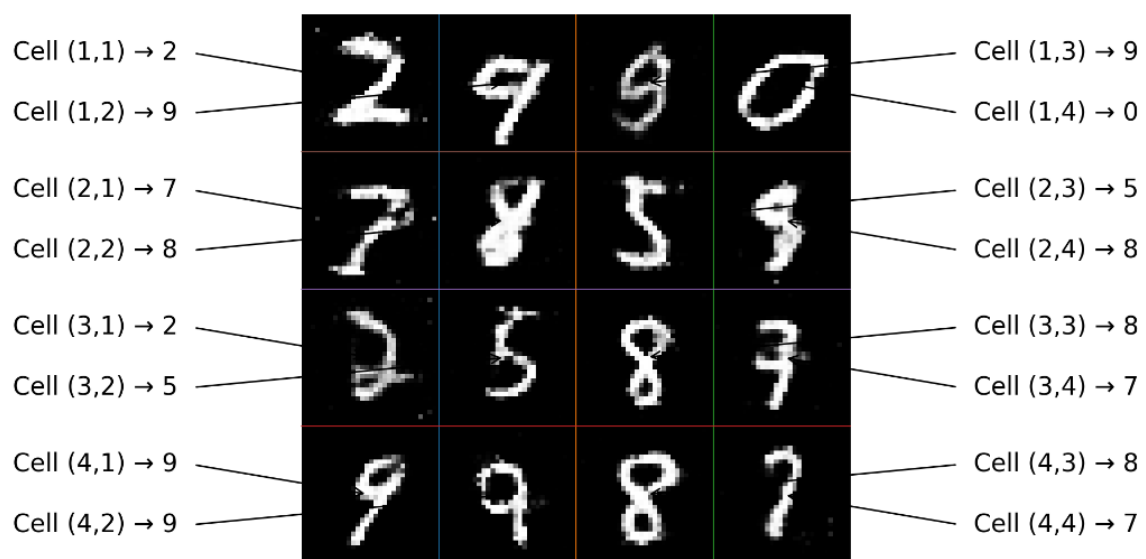


Epoch 50: Final trained output showing mature digit formation and class variation.

D Loss = 1.0710 | G Loss = 1.0218

GAN output quality improved progressively across epochs, while adversarial losses fluctuated due to the competitive learning process between generator and discriminator.

GAN Generated Handwritten Digit Interpretation



4x4 synthetic sample grid from trained GAN generator

- Each cell is generated from random latent noise
- Digits are synthetic outputs from trained generator
- Variation reflects learned handwritten digit distribution